MEDIUM ALGEBRA ANSWER KEY

Algebra

06

10 answers · Rationale-rich · Teach from doc

Q1 C

C. $8r + 18t = 210$

Choice C is correct. It's given that a total of 2 squares each have side length r. Therefore, each of the squares has perimeter $4r$. Since there are a total of 2 squares, the sum of the perimeters of these squares is $4r + 4r$, which is equivalent to $24r$, or $8r$. It's also given that a total of 6 equilateral triangles each have side length t. Therefore, each of the equilateral triangles has perimeter $3t$. Since there are a total of 6 equilateral triangles, the sum of the perimeters of these triangles is $3t + 3t + 3t + 3t + 3t + 3t$, which is equivalent to $63t$, or $18t$. Since the sum of the perimeters of the squares is $8r$ and the sum of the perimeters of the triangles is $18t$, the sum of the perimeters of all these squares and triangles is $8r + 18t$. It's given that the sum of the perimeters of all these squares and triangles is 210. Therefore, the equation $8r + 18t = 210$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2 C

C. $950 + 50(t - 3) = 1{,}150$

Choice C is correct. It's given that the cost to rent a bus is $\$950$ for the first 3 hours and an additional $\$50$ per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t - 3$. The cost for these additional hours is $\$50$ per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is $\$1{,}150$. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1{,}150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q3

C

C. $fx = 4x + 2$

Choice C is correct. In the xy-plane, the graph of a linear function can be written in the form $fx = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the graph of $y = fx$. It's given that the graph of the linear function f, where $y = fx$, in the xy-plane contains the point $0, 2$. Thus, $b = 2$. The slope of the graph of a line containing any two points x_1, y_1 and x_2, y_2 can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $0, 2$ and $8, 34$, it follows that the slope of the graph of the line containing these points is $m = \frac{34 - 2}{8 - 0}$, or $m = 4$. Substituting 4 for m and 2 for b in $fx = mx + b$ yields $fx = 4x + 2$.

Choice A is incorrect. This function represents a graph with a slope of 2 and a y-intercept of $0, 42$.

Choice B is incorrect. This function represents a graph with a slope of 32 and a y-intercept of $0, 36$.

Choice D is incorrect. This function represents a graph with a slope of 8 and a y-intercept of $0, 2$.

Q4

D

D. 15

Choice D is correct. Multiplying both sides of $y = 2x - 3$ by 5 results in $5y = 10x - 15$. Multiplying both sides of $3y = 5x$ by 2 results in $6y = 10x$. Subtracting the resulting equations yields $5y - 6y = (10x - 15) - (10x)$, which simplifies to $-y = -15$. Dividing both sides of $-y = -15$ by -1 results in $y = 15$.

Choices A and B are incorrect and may result from incorrectly subtracting the transformed equation. Choice C is incorrect and may result from finding the value of x instead of the value of y.

Q5

25

The correct answer is 25. The total cost of the party is found by adding the onetime fee of the venue to the cost per attendee times the number of attendees. Let x be the number of attendees. The expression $35 + 10.25x$ thus represents the total cost of the party. It's given that the budget is $\$300$, so this situation can be represented by the inequality $35 + 10.25x \leq 300$. Subtracting 35 from both sides of this inequality gives $10.25x \leq 265$. Dividing both sides of this inequality by 10.25 results in approximately $x \leq 25.854$. Since the question is stated in terms of attendees, rounding 25.854 down to the greatest whole number gives the greatest number of attendees possible, which is 25.

Q6

B

B. $8x + 12y = 480$

Choice B is correct. The graph shown is a line passing through the points $0, 40$ and $60, 0$. Since the relationship between x and y is linear, if two points on the graph make a linear equation true, then the equation represents the relationship. Substituting 0 for x and 40 for y in the equation in choice B, $8x + 12y = 480$, yields $80 + 1240 = 480$, or $480 = 480$, which is true. Substituting 60 for x and 0 for y in the equation $8x + 12y = 480$ yields $860 + 120 = 480$, or $480 = 480$, which is true. Therefore, the equation $8x + 12y = 480$ represents the relationship between x and y.

Choice A is incorrect. The point $0, 40$ is not on the graph of this equation, since $40 = 80 + 12$, or $40 = 12$, is not true.

Choice C is incorrect. The point $0, 40$ is not on the graph of this equation, since $40 = 120 + 8$, or $40 = 8$, is not true.

Choice D is incorrect. The point $0, 40$ is not on the graph of this equation, since $120 + 840 = 480$, or $320 = 480$, is not true.

Q7

D

D. -10

Choice D is correct. Multiplying the given equation by 2 on each side yields $2\left(4x-\frac{1}{2}\right)=2(-5)$. Applying the distributive property, this equation can be rewritten as $2(4x)-2\left(\frac{1}{2}\right)=2(-5)$, or $8x-1=-10$.

Choices A, B, and C are incorrect and may result from calculation errors in solving the given equation for x and then substituting that value of x in the expression $8x-1$.

Q8

A

A. $P(t)=862-1.6t$

Choice A is correct. It is given that the relationship between population and year is linear; therefore, the function that models the population t years after 2000 is of the form $P(t)=mt+b$, where m is the slope and b is the population when $t=0$. In the year 2000, $t=0$. Therefore, $b=862$. The slope is given by $m=\frac{P(10)-P(0)}{10-0}=\frac{846-862}{10-0}=\frac{-16}{10}=-1.6$. Therefore, $P(t)=-1.6t+862$, which is equivalent to the equation in choice A.

Choice B is incorrect and may be the result of incorrectly calculating the slope as just the change in the value of P. Choice C is incorrect and may be the result of the same error as in choice B, in addition to incorrectly using t to represent the year, instead of the number of years after 2000. Choice D is incorrect and may be the result of incorrectly using t to represent the year instead of the number of years after 2000.

Q9

A

A. Zero

Choice A is correct. A system of two linear equations in two variables, x and y, has zero points of intersection if the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the y-coordinates of their y-intercepts, b, are different and are parallel if their slopes, m, are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1, respectively, and the values of m are both 2. Since the values of b are different, the graphs of these lines have different y-coordinates of the y-intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the xy-plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and the same y-coordinate of the y-intercept. Since the given equations represent lines with different y-coordinates of their y-intercepts, there are not infinitely many intersection points.

Q10 A

A. $2x + 100 \leq 210$

Choice A is correct. The perimeter of a rectangle is equal to the sum of 2 times its length and 2 times its width. It's given that the rectangle's length is 50 inches and the width is x inches. Therefore, the perimeter, in inches, is $250 + 2x$, or $100 + 2x$, which is equivalent to $2x + 100$. It's given that the perimeter is at most 210 inches; therefore, $2x + 100 \leq 210$ represents this situation.

Choice B is incorrect. This inequality represents a situation where the perimeter is at least, rather than at most, 210 inches.

Choice C is incorrect. This inequality represents a situation where 2 times the length, rather than the length, is 50 inches.

Choice D is incorrect. This inequality represents a situation where 2 times the length, rather than the length, is 50 inches, and the perimeter is at least, rather than at most, 210 inches.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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